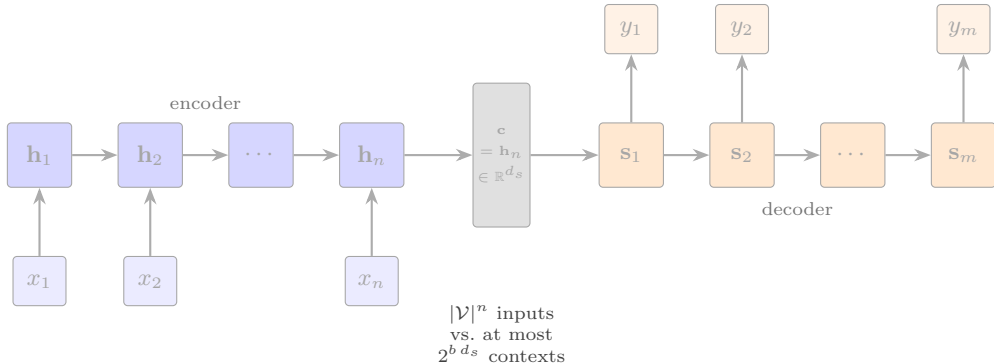




Every path from input to output passes through the single vector $\mathbf{c}$: its dimension d_s is fixed while the input alphabet $|\mathcal{V}|^n$ grows without bound — a pigeonhole bottleneck, not a modeling choice.